

ML/DL Ops Assignment 3: Short Report

End-to-End Hugging Face Model Training & Docker Deployment

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1 Project Overview

This report details the end-to-end machine learning pipeline developed for Assignment 3. The objective was to build a complete ML workflow: preparing a dataset, fine-tuning a pre-trained model using the Hugging Face ecosystem, containerizing the evaluation process via Docker, and publishing all artifacts to GitHub and the Hugging Face Hub.

2 Dataset & Preprocessing

Dataset Used: Goodreads Book Reviews.

Given the massive scale of the original `.json.gz` files (over 14 million records), a random sample of 15,000 records was selected to ensure efficient training while maintaining statistical significance.

- **Label Mapping:** Original 1–5 star ratings were converted into binary sentiment labels. Ratings strictly greater than 3 were mapped to 1 (Positive), while ratings of 3 or below were mapped to 0 (Negative).
- **Tokenization:** The text data was tokenized using the `distilbert-base-uncased` tokenizer with a maximum sequence length of 128 to optimize memory usage and processing speed.

3 Model Selection & Training

Model Selected: DistilBERT (`distilbert-base-uncased`)

Rationale: DistilBERT was chosen for its optimal balance of predictive performance and computational efficiency. It retains over 95% of BERT’s language understanding capabilities while being significantly smaller and faster, making it highly suitable for containerized deployment.

3.1 Training Configuration

The model was fine-tuned using the Hugging Face `Trainer` API with the following key arguments:

- **Epochs:** 2
- **Learning Rate:** 2×10^{-5}
- **Batch Size:** 16 (per device)
- **Hardware Acceleration:** Training was accelerated locally utilizing the Apple Metal Performance Shaders (MPS) backend on the M4 chip, allowing the 2-epoch training to complete in approximately 9.5 minutes.

4 Evaluation & Deployment Metrics

To guarantee deployment reliability, the model was evaluated in two isolated environments. The results match perfectly, confirming successful remote deployment.

Metric	Local Evaluation	HF Hub via Docker
Accuracy	78.17% (0.7816)	78.17% (0.7816)
F1 Score	0.8458 (0.8457)	0.8458 (0.8457)
Evaluation Loss	0.4522	0.4522
Runtime	~18 seconds (MPS)	~86 seconds (CPU)

Table 1: Comparison of evaluation metrics across environments.

Conclusion: The precise match in accuracy, F1 score, and loss between the local script and the Dockerized container (which pulled the model directly from the Hugging Face Hub) conclusively verifies that the model was correctly published and functions seamlessly in a production-ready setup.

5 Challenges & Solutions

- **Docker Image Optimization & Large Datasets:** The most significant challenge was handling the massive Goodreads dataset within Docker. Copying the data directly into the image would have bloated it to several gigabytes. This was resolved by implementing **Docker Volume Mounting** (`-v`), which allowed the lightweight evaluation container to read the dataset directly from the local host directory.
- **Library Version Dependencies:** An unexpected keyword argument `'evaluation_strategy'` error occurred due to recent deprecations in the Hugging Face `transformers` library. The code was updated to use the modernized `eval_strategy` parameter, ensuring compatibility with the latest API standards.

6 Repository Links

- **GitHub Repository:** <https://github.com/m25csa003-glitch/mlops-hf-docker-assignment>
- **Hugging Face Model Hub:** <https://huggingface.co/Despo08/distilbert-goodreads-assignment>